Math GRE Prep Day 5

W.R. Casper

Department of Mathematics
California State University Fullerton

March 14, 2025

Outline

Outline

Problem 1

Find the Jacobian of the transformation

$$\begin{cases} u &= xe^{xy} \\ v &= ye^{xy} \end{cases}$$

Problem 2

Let $f(x, y) = x^3 - axy + y^2 - x$. For what values of a does the function have a local minimum?

Problem 3

Find

$$\iint_R z dx dy$$

where $z = 8xy$ and R is the part of the unit disk in the first quadrant.

Problem 4

Find the length of the parametric curve $x = \cos(t)e^t$ and $y = -\sin(t)e^t$ between $0 < t < 1$.

Problem 5

If $\mathbf{F} = \langle 2y - 2x, x^2 + y \rangle$, find the value of $\int_C \mathbf{F} \cdot d\mathbf{r}$ where C is the segment of the parabola $y = x^2$ starting from $(-1, 1)$ and going to $(0, 0)$.

Problem 6

Let C be the portion of the astroid $x^{2/3} + y^{2/3} = 1$ from $(1, 0)$ to $(0, 1)$, which is parameterized by the equations

$$x = \cos^3 t, \quad y = \sin^3 t, \; 0 \le t \le \pi/2.$$

Evaluate the integral

$$\int_C (y\cos(xy) - 1)dx + (1 + x\cos(xy))dy$$

Problem 7

Find all functions $f(x, y)$ satisfying $\frac{\partial f}{\partial x} = 2x + y$ and $\frac{\partial f}{\partial x} = x + 2y$.

Problem 8

Find the point on the plane $2x + y + 3z = 3$ which is closest to the origin.

Problem 9

Set up an integral which represents the volume of the solid bounded above by the graph of $z = 6 - x^2 - 2y^2$ and below by the graph of $z = -2 + x^2 + 2y^2$.

Problem 10

Minimize the function $f(x, y, z) = x + 4z$ on the surface $x^2 + y^2 + z^2 = 1$.